

3.7.3 Electric fields (A-level only)

3.7.3.1 Coulomb's law (A-level only)

Content	Opportunities for skills development
Force between point charges in a vacuum: $F = \frac{1}{4\pi\epsilon_0} \frac{Q_1 Q_2}{r^2}$ Permittivity of free space, ϵ_0 Appreciation that air can be treated as a vacuum when calculating force between charges. For a charged sphere, charge may be considered to be at the centre. Comparison of magnitude of gravitational and electrostatic forces between subatomic particles.	MS 0.3, 2.3 Students can estimate the magnitude of the electrostatic force between various charge configurations.

3.7.3.2 Electric field strength (A-level only)

Content	Opportunities for skills development
Representation of electric fields by electric field lines. Electric field strength. E as force per unit charge defined by $E = \frac{F}{Q}$ Magnitude of E in a uniform field given by $E = \frac{V}{d}$ Derivation from work done moving charge between plates: $Fd = Q\Delta V$ Trajectory of moving charged particle entering a uniform electric field initially at right angles. Magnitude of E in a radial field given by $E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2}$	PS 1.2, 2.2 / AT b Students can investigate the patterns of various field configurations using conducting paper (2D) or electrolytic tank (3D).

3.7.3.3 Electric potential (A-level only)

Content	Opportunities for skills development
Understanding of definition of absolute electric potential, including zero value at infinity, and of electric potential difference. Work done in moving charge Q given by $\Delta W = Q \Delta V$ Equipotential surfaces. No work done moving charge along an equipotential surface. Magnitude of V in a radial field given by $V = \frac{1}{4\pi\epsilon_0} \frac{Q}{r}$ Graphical representations of variations of E and V with r. V related to E by $E = \frac{\Delta V}{\Delta r}$ ΔV from the area under graph of E against r.	